Hybrid Quantum-Classical Framework for Woodland Segmentation and QUBO-Based Region Optimization

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Abstract**—***Remote sensing image segmentation has become an important tool for environmental monitoring, woodland analysis, and automated land-cover mapping. Classical deep learning models can produce useful segmentation masks, yet post-processing still becomes difficult when the model generates overlapping candidate regions or uncertain region proposals. This paper presents a hybrid quantum-classical framework for woodland segmentation and QUBO-based region optimization. The proposed pipeline combines an Attention U-Net segmentation backbone, a simulated variational quantum bottleneck, and a Quadratic Unconstrained Binary Optimization formulation for selecting non-redundant woodland regions. The work is motivated by recent quantum-assisted segmentation research and quantum-ready QUBO suppression methods. The implementation uses PyTorch for the classical network and PennyLane for the simulated quantum layer. The resulting framework provides a practical research prototype for studying how near-term quantum ideas may support remote sensing workflows, especially where segmentation and combinatorial selection are tightly connected.*

Keywords**—***Attention U-Net, QUBO, quantum machine learning, remote sensing, woodland segmentation, variational quantum circuits*

I. INTRODUCTION

Remote sensing imagery allows researchers and agencies to monitor vegetation, urbanization, forest cover, and environmental change at a large spatial scale. A common requirement in such workflows is to separate the target land-cover class from the surrounding background. For woodland mapping, this means identifying connected vegetation regions with enough spatial reliability to be useful for downstream decision-making.

Convolutional neural networks, especially U-Net based encoder-decoder models, are widely used for image segmentation because they preserve spatial detail through skip connections while learning hierarchical semantic features. Attention U-Net further improves this design by using attention gates to emphasize relevant regions and suppress weak or irrelevant activations. However, segmentation alone does not always solve the full mapping problem. The predicted mask often contains broken components, overlapping candidate boxes, and redundant proposals that must be filtered after inference.

Quantum computing is being explored for two related reasons. First, variational quantum circuits can be inserted into classical machine learning pipelines as feature transformation modules. Second, many post-processing problems can be written as combinatorial optimization tasks and expressed as QUBO problems. These properties make hybrid quantum-classical systems a natural fit for experiments that combine segmentation with region selection.

This paper presents a student research prototype that integrates these ideas into a single workflow. The proposed framework uses Attention U-Net for woodland segmentation, a variational quantum bottleneck for feature transformation, and QUBO-based optimization for candidate region selection. The goal is not to claim quantum advantage on current hardware, but to design a quantum-ready formulation that can be evaluated on simulators today and adapted to quantum solvers in future work.

II. RELATED WORK

A. Quantum-Assisted Remote Sensing Segmentation

Recent work on quantum-assisted remote sensing has shown that quantum preprocessing can be connected to classical segmentation models. Russo et al. investigated a Quantum-assisted Attention U-Net for building segmentation using Sentinel-1 SAR data over Tunis. Their method used quanvolutional preprocessing with angle encoding to generate feature maps before feeding them to an Attention U-Net. The study reported that quantum-assisted variants preserved competitive overall accuracy while greatly reducing the number of trainable parameters compared with the classical Attention U-Net baseline [1].

This idea is relevant to woodland segmentation because both tasks require pixel-wise interpretation of remote sensing imagery. In both cases, the segmentation model must preserve local spatial structure while learning broader patterns from the scene.

B. QUBO-Based Suppression and Region Selection

QUBO-based suppression has recently been explored as an alternative to greedy Non-Maximum Suppression in object detection. Yamamura et al. proposed quantum-ready QUBO formulations that include appearance and confidence features to distinguish redundant detections from overlapping true objects [2]. Their work demonstrated that QUBO formulations can improve suppression quality, especially in crowded scenes where greedy suppression may remove valid detections.

Although the present project focuses on woodland segmentation rather than object detection, the optimization challenge is similar: a model may produce several candidate regions that overlap or represent the same underlying area. QUBO provides a structured way to reward high-confidence regions and penalize redundant selections.

III. PROPOSED METHODOLOGY

A. Overall Pipeline

The proposed workflow is divided into five main stages: image preprocessing, Attention U-Net segmentation, quantum bottleneck transformation, candidate region extraction, and QUBO-based candidate selection. Fig. 1 summarizes the overall flow of the system.

Input image
Attention U-Net + quantum bottleneck
Prediction mask → candidates → QUBO selection

Fig. 1. Proposed hybrid quantum-classical pipeline for woodland segmentation and QUBO-based region optimization.

B. Attention U-Net Backbone

The segmentation stage uses Attention U-Net as the classical backbone. The encoder extracts feature maps at multiple spatial resolutions, while the decoder reconstructs the output mask. Skip connections transfer spatial detail from the encoder to the decoder. Attention gates are applied to these skip connections so that the model can assign greater importance to woodland-related features.

Let x_i represent encoder features and g represent decoder gating features. The simplified attention operation can be written as

$$q = RELU(W_x x_i + W_g g + b)$$

$$alpha = sigmoid(W_{psi} q)$$

$$x_{out} = alpha * x_i$$

Here, alpha acts as the attention coefficient map. Features with higher attention are passed more strongly to the decoder, while weaker activations are suppressed.

C. Quantum Bottleneck Layer

A variational quantum circuit is inserted at the bottleneck of the segmentation network. The bottleneck is selected because it contains compressed semantic information and is therefore a practical location for quantum feature transformation. Before entering the quantum layer, the classical feature vector is reduced to match the number of available simulated qubits.

Classical features are encoded through angle encoding. For an input vector x, the feature map prepares a quantum state using parameterized rotations. A simplified representation is

$$U_\phi |0>^n = |\phi(x)>$$

The circuit then applies trainable rotation gates and entangling layers. Measurement results are converted back into classical values and passed to the decoder. In the current implementation, this layer is executed using a simulator rather than real quantum hardware.

D. Candidate Region Generation

After inference, the predicted probability mask is thresholded to obtain a binary woodland mask. Connected components and contour extraction are then used to identify candidate regions. Each candidate is represented by a bounding box and a score derived from the predicted mask confidence, region size, and local consistency.

This stage intentionally produces more candidates than needed. The role of the optimization stage is to select the most useful subset while discouraging redundant overlap.

E. QUBO-Based Region Optimization

The candidate selection problem is formulated as a QUBO problem. Let x_i be a binary variable where $x_i = 1$ means that candidate i is selected and $x_i = 0$ means it is rejected. The QUBO objective is

$$\text{maximize} \ \ x^t * Q * x \ \text{ where } \ x \text{ in } \{0, 1\}^n .$$

The coefficient matrix Q contains positive diagonal terms for candidate confidence and negative off-diagonal terms for pairwise overlap penalties. A simplified formulation is

$$Q = w_1 L - w_2 P_{overlap} + w_3 P_{spatial}$$

L is the diagonal confidence matrix, P_overlap penalizes candidate pairs with excessive intersection, and P_spatial measures spatial redundancy. The weights w1, w1, and w3 control the balance between selecting confident regions and suppressing duplicate regions.

IV. IMPLEMENTATION

A. Software Stack

The implementation was developed in Python. PyTorch was used for neural network construction and training, PennyLane was used for the variational quantum layer, OpenCV was used for contour extraction and region processing, and NumPy was used for numerical matrix operations. The experiments were organized in a Jupyter notebook to support iterative testing and visualization.

TABLE I. MAIN COMPONENTS OF THE PROPOSED FRAMEWORK

Purpose	To
Woodland mask	PyTo
Feature transform	Penny
Region proposals	Oper
Final selection	Matrix

B. Training and Inference Flow

During training, images and masks are passed through the hybrid segmentation model. The loss function combines binary cross-entropy with Dice-based overlap loss to encourage both pixel-wise correctness and region-level consistency. During inference, the model predicts woodland probability masks. These masks are post-processed into candidate boxes and optimized through the QUBO module.

C. Solver Strategy

The QUBO formulation is quantum-ready, but the current implementation uses classical or simulated solvers because reliable large-scale quantum hardware is not yet available for this project setting. This mirrors the practical approach used in recent QUBO suppression research, where classical solvers are used to evaluate formulations before deployment on future quantum systems.

V. RESULTS AND DISCUSSION

The implemented framework successfully produced woodland segmentation masks and converted those masks into candidate regions. The Attention U-Net backbone provided a stable segmentation baseline, while the quantum bottleneck introduced an experimental feature transformation stage. The QUBO module reduced redundant region proposals by balancing confidence rewards with overlap penalties.

The most important observation from this project is that segmentation and candidate selection should not be treated as completely separate problems. A good segmentation mask may still require careful region-level optimization if the final output is expected to contain selected regions rather than only pixels. QUBO provides a clean mathematical structure for this selection step.

The project also showed that quantum-assisted ideas can be integrated into a classical deep learning workflow without replacing the entire model. This is important because current NISQ-era quantum devices are limited in qubit count, noise tolerance, and circuit depth. A hybrid design allows the classical model to handle most of the spatial learning while the quantum layer and QUBO formulation explore targeted parts of the pipeline.

TABLE II. QUALITATIVE EFFECT OF MAJOR MODULES

dule	Benefit
ntion	Focus regions
ntum	Feature transform
JBO	Reduce duplicates

VI. LIMITATIONS

The current system is a research prototype and therefore has several limitations. First, the quantum circuit is executed on a simulator, which means the study does not demonstrate hardware-level quantum advantage. Second, the quantum bottleneck is constrained by the number of simulated qubits and the size of the feature vector that can be encoded. As the feature dimension increases, the circuit requirements also grow, limiting scalability.

Third, the complexity of the QUBO formulation increases with the number of candidate regions. Therefore, very large scenes may require candidate filtering or preprocessing before optimization in order to keep the problem computationally feasible. Fourth, the datasets used in the referred papers were behind a paywall, so we could not conduct a complete end-to-end comparison between our approach and the original methods. This limitation explains the absence of direct result comparison, as our work primarily introduces a proof-of-concept research prototype rather than a fully benchmarked production-level system.

Another limitation is that the final detection results depend heavily on thresholding and connected-component extraction. Improved candidate generation strategies could produce higher-quality QUBO inputs, reduce unnecessary solver workload, and potentially improve the overall performance of the proposed approach.

VII. FUTURE WORK

Future extensions can improve the framework in several directions. The segmentation model can be trained and evaluated on larger remote sensing datasets with stronger quantitative metrics such as Intersection over Union, Dice score, precision, and recall. The QUBO formulation can be extended to include appearance similarity, confidence-aware pairwise terms, and multi-scale spatial penalties. The quantum circuit design can also be compared across different ansatz structures, including strongly entangling layers and random circuit variants.

A practical next step is to evaluate whether the QUBO module can be replaced by QAOA-inspired solvers or quantum annealing backends when hardware access is available. This would allow a clearer comparison between classical optimization and quantum-ready optimization in the same woodland mapping pipeline.

VIII. CONCLUSION

This paper presented a hybrid quantum-classical framework for woodland segmentation and QUBO-based region optimization. The system combines Attention U-Net, a variational quantum bottleneck, and QUBO-based candidate selection into a single remote sensing workflow. The work demonstrates that quantum-inspired methods can be meaningfully connected to practical segmentation pipelines, especially when the task includes both pixel-level prediction and region-level optimization. While the current implementation relies on simulation, it provides a foundation for future experiments on quantum-ready remote sensing and combinatorial optimization systems.

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